Completion, Uniform Structure, and Quotient Topology

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§1.1 Cauchy Sequences and Missing Points

A sequence (x_n) in a metric space (X, d) is Cauchy when for every $\varepsilon > 0$ there is N such that

$$d(x_m, x_n) < \varepsilon \qquad (m, n \geq N).$$

It records points that eventually become mutually indistinguishable at every metric scale. A complete space contains a limit for every such sequence.

The rational sequence

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n}$$

is Cauchy in $\mathbb{Q}$, but it has no rational limit. Nothing is wrong with the sequence; the intended limit is absent from the space. Completion constructs a space in which every coherent approximation process has a point to name.

Two Cauchy sequences should represent the same new point when their mutual distance tends to zero:

$$(x_n) \sim (y_n) \quad \Longleftrightarrow \quad d(x_n, y_n) \longrightarrow 0.$$

Reflexivity and symmetry are immediate. If $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$, then

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n) \longrightarrow 0,$$

so the relation is transitive.

§1.2 The completion metric is a limit of mutual distances

Let $\widehat{X}$ be the set of equivalence classes of Cauchy sequences. Define

$$\widehat{d}([(x_n)], [(y_n)]) = \lim_{n \to \infty} d(x_n, y_n).$$

The scalar sequence on the right is Cauchy because

$$|d(x_n, y_n) - d(x_m, y_m)| \leq d(x_n, x_m) + d(y_n, y_m).$$

Hence the limit exists in $\mathbb{R}$.

The value does not depend on the chosen representatives. If

$$(x_n) \sim (x'_n), \qquad (y_n) \sim (y'_n),$$

then

$$|d(x_n, y_n) - d(x'_n, y'_n)| \leq d(x_n, x'_n) + d(y_n, y'_n) \longrightarrow 0.$$

Thus both representative pairs give the same limit.

Nonnegativity and symmetry pass directly to the limit. If $\widehat{d} = 0$, then $d(x_n, y_n) \to 0$, so the classes are equal. Finally,

$$d(x_n, z_n) \leq d(x_n, y_n) + d(y_n, z_n)$$

passes to the limit and gives the triangle inequality. Therefore $\widehat{d}$ is a genuine metric.

§1.3 Embedding into the Completion

Send $x \in X$ to the constant sequence:

$$\iota(x) = [(x, x, x, \ldots)].$$

Then

$$\widehat{d}(\iota(x), \iota(y)) = d(x, y),$$

so ι is an isometric embedding.

Its image is dense. Let

$$\xi = [(x_n)] \in \widehat{X}.$$

For fixed N,

$$\widehat{d}(\xi, \iota(x_N)) = \lim_{m \to \infty} d(x_m, x_N).$$

Because (x_n) is Cauchy, this quantity tends to zero as $N \to \infty$. Thus

$$\iota(x_N) \longrightarrow \xi.$$

Every completed point is a limit of original points.

For $X = \mathbb{Q}$, the decimal truncations (q_n) of $\sqrt{2}$ define a class in $\widehat{\mathbb{Q}}$. It cannot equal any constant rational class: if it equalled $r \in \mathbb{Q}$, then

$$|q_n - r| \longrightarrow 0,$$

forcing $r = \sqrt{2}$. The new class is exactly the missing irrational point described by rational approximations.

§1.4 Completeness by a Diagonal Argument

Let (ξ_r) be a Cauchy sequence in $\widehat{X}$. Passing to a subsequence if necessary, arrange that

$$\widehat{d}(\xi_r, \xi_{r+1}) < 2^{-r}.$$

Density of $\iota(X)$ lets us choose $y_r \in X$ with

$$\widehat{d}(\iota(y_r), \xi_r) < 2^{-r}.$$

Then

$$\begin{aligned} d(y_r, y_{r+1}) &\leq \widehat{d}(\iota(y_r), \xi_r) + \widehat{d}(\xi_r, \xi_{r+1}) \\ &\quad + \widehat{d}(\xi_{r+1}, \iota(y_{r+1})) \\ &< 2^{-r} + 2^{-r} + 2^{-(r+1)}. \end{aligned}$$

The geometric tail shows that (y_r) is Cauchy in X. Let

$$\xi = [(y_r)] \in \widehat{X}.$$

The construction gives

$$\xi_r \longrightarrow \xi$$

along the chosen subsequence, and since the original sequence was Cauchy, the whole sequence has the same limit. Hence $\widehat{X}$ is complete.

The argument exposes the mechanism: approximate each abstract completed point by an original point with rapidly shrinking error, then use one diagonal Cauchy sequence in X to encode the limit.

§1.5 Extending Uniformly Continuous Maps

Let Y be complete and let

$$f : X \to Y$$

be uniformly continuous. If (x_n) is Cauchy, then $(f(x_n))$ is Cauchy. Indeed, for every $\varepsilon > 0$, choose one $\delta > 0$ working simultaneously for all points:

$$d_X(x, x') < \delta \quad \Longrightarrow \quad d_Y(f(x), f(x')) < \varepsilon.$$

The Cauchy condition eventually places every pair x_m, x_n within δ.

Define

$$\widehat{f} : \widehat{X} \to Y, \qquad \widehat{f}([(x_n)]) = \lim_{n\to\infty} f(x_n).$$

If $(x_n) \sim (y_n)$, uniform continuity gives

$$d_Y(f(x_n), f(y_n)) \longrightarrow 0,$$

so the definition is independent of representatives. It extends f because constant sequences retain their values.

The extension is unique. If $g : \widehat{X} \to Y$ is continuous and agrees with f on X, then for

$$\xi = \lim_n \iota(x_n)$$

one has

$$g(\xi) = \lim_n g(\iota(x_n)) = \lim_n f(x_n) = \widehat{f}(\xi).$$

Thus completion is characterized by the extension property

uniformly continuous maps into complete spaces extend uniquely.

Uniform continuity is essential. The continuous map

$$f : (0, 1) \to \mathbb{R}, \qquad f(x) = \frac{1}{x}$$

has no continuous extension to the completion $[0, 1]$, because it diverges along $x_n = 1/n \to 0$.

§1.6 One topology, two different completions

A homeomorphism need not preserve Cauchy sequences. In fact, the same underlying set can carry two metrics with exactly the same open sets and different completions. On $\mathbb{R}$, compare

$$d_{\mathrm{E}}(x, y) = |x - y|, \qquad d_{\mathrm{A}}(x, y) = |\arctan x - \arctan y|.$$

The strictly increasing homeomorphism $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ shows that d_{A} induces the usual topology on $\mathbb{R}$. Yet the sequence $x_n = n$ is not Cauchy for d_{E}, while it is Cauchy for d_{A}, because

$$|\arctan n - \arctan m| \longrightarrow 0 \quad (n, m \to \infty).$$

Similarly $-n$ is d_{A}-Cauchy. The isometry furnished by arctangent identifies $(\mathbb{R}, d_{\mathrm{A}})$ with the open interval $(-\pi/2, \pi/2)$ with its usual metric. Thus

$$\widehat{(\mathbb{R}, d_{\mathrm{A}})} \cong [-\pi/2, \pi/2],$$

whereas $(\mathbb{R}, d_{\mathrm{E}})$ is already complete. The first completion adds two endpoints and the second adds nothing.

The distinction does not come from pointwise convergence to existing points: the two metrics agree about that. It comes from whether *pairs of points far out in the space* become mutually close. A homeomorphism preserves local neighborhoods but not uniform closeness. This is exactly the extra information encoded by a uniform structure.

§1.7 The universal property makes completion unique

Suppose

$$\iota_X : X \hookrightarrow Y, \qquad j_X : X \hookrightarrow Z$$

are dense isometric embeddings into complete metric spaces. Apply the extension property to the uniformly continuous map

$$j_X \circ \iota_X^{-1} : \iota_X(X) \to Z.$$

It extends uniquely to an isometry

$$F : Y \to Z.$$

Reversing the roles gives

$$G : Z \to Y.$$

Both composites agree with the identity on the dense copy of X. By uniqueness of extension,

$$GF = \mathrm{id}_Y, \qquad FG = \mathrm{id}_Z .$$

Therefore any two completions are uniquely isometric in a way that fixes X.

The quotient of Cauchy sequences is one concrete construction; the universal property is what makes the result canonical.

§1.8 Homeomorphism forgets the uniform scale of closeness

The interval $(0, 1)$ and the real line are homeomorphic via

$$h(x) = \tan\bigl(\pi(x - 1/2)\bigr).$$

Yet $(0, 1)$ is bounded and incomplete in its Euclidean metric, while $\mathbb{R}$ is unbounded and complete. Boundedness and completeness are therefore not topological invariants.

The map is not uniformly continuous. Take

$$x_n = 1 - \frac{1}{n}, \qquad y_n = 1 - \frac{2}{n}.$$

Then

$$|x_n - y_n| = \frac{1}{n} \longrightarrow 0,$$

but

$$h(x_n) = \cot\frac{\pi}{n}, \qquad h(y_n) = \cot\frac{2\pi}{n},$$

and their difference grows on the order of n. No single δ controls the map near both endpoints.

Topology records which sets are neighborhoods. Uniform structure additionally records which pairs are uniformly close across the whole space. Completion depends on the latter, because Cauchy sequences are uniform objects.

§1.9 When a Bijection Is a Homeomorphism

The map

$$f : [0, 2\pi) \to S^1, \qquad f(t) = (\cos t, \sin t)$$

is continuous and bijective, but its inverse is not continuous at $(1, 0)$. Points approaching $(1, 0)$ from the lower semicircle have parameters tending to 2π, while the point itself has parameter 0.

A useful positive theorem is the compact–Hausdorff criterion.

Theorem 1.9.1

If X is compact, Y is Hausdorff, and $f : X \to Y$ is a continuous bijection, then f is a homeomorphism.

Proof. A closed subset $C \subseteq X$ is compact. Its image $f(C)$ is compact in Y, hence closed because Y is Hausdorff. Thus f is a closed map. For any closed $C \subseteq X$,

$$(f^{-1})^{-1}(C) = f(C)$$

is closed, so f^{-1} is continuous. □

The failed half-open-interval example lacks compactness. The theorem identifies exactly which global hypotheses make inverse continuity automatic.

§1.10 Quotient Topology and Gluing

Let $q : X \to Y$ be a surjection. The quotient topology on Y is defined by

$$U \subseteq Y \text{ open} \quad \Longleftrightarrow \quad q^{-1}(U) \text{ open in } X.$$

It is the strongest topology on Y making q continuous.

Its universal property is the factorization test:

$$g : Y \to Z \text{ is continuous} \quad \Longleftrightarrow \quad g \circ q : X \to Z \text{ is continuous.}$$

Thus a map out of a quotient is constructed upstairs, where the original points are still visible, provided the map is constant on equivalence classes.

A subset $A \subseteq X$ is saturated when it is a union of equivalence classes:

$$A = q^{-1}(q(A)).$$

For arbitrary A,

$$q(A) \text{ is open in } Y \quad \Longleftrightarrow \quad q^{-1}(q(A)) \text{ is open in } X.$$

A quotient map need not send every open set to an open set. On $[0, 1]$, collapse the entire interval $[0, 1/2]$ to one point. The open set $(0, 1/2)$ has saturation $[0, 1/2]$, which is not open, so its image is not open in the quotient.

The quotient topology controls precisely the saturated information that survives the identification.

§1.11 A circle can be made by completing rational angles

Completion and quotient topology meet very naturally in the simplest periodic space. Consider the additive quotient group

$$X = \mathbb{Q}/\mathbb{Z}.$$

Here $[q]$ means that q and $q+k$ represent the same point for every integer k. The usual metric $|q-r|$ cannot simply be written on quotient classes, because it depends on which representatives we pick. The right metric is the circular distance

$$\boxed{d_X([q],[r]) = \min_{k\in\mathbb{Z}} |q-r-k|.}$$

Replacing either representative by an integer translate merely reindexes the minimum. Also $d_X = 0$ exactly when $q - r \in \mathbb{Z}$, so it is a genuine metric. The triangle inequality follows by choosing integer translates for each of the two legs and adding their distances.

The map

$$\iota : \mathbb{Q}/\mathbb{Z} \longrightarrow S^1, \qquad [q] \longmapsto e^{2\pi i q}$$

is well defined and injective. Equip S^1 with the angular metric

$$d_{S^1}(e^{2\pi i x}, e^{2\pi i y}) = \min_{k\in\mathbb{Z}} |x-y-k|.$$

By construction, ι is an *isometry*. Rational angles are dense in all real angles, so its image is dense in S^1. The circle with its angular metric is compact and hence complete. By the uniqueness of metric completion,

$$\boxed{\widehat{\mathbb{Q}/\mathbb{Z}} \cong S^1.}$$

An actual missing point is easy to exhibit. Let

$$q_n = \frac{\lfloor 10^n \sqrt{2} \rfloor}{10^n} \in \mathbb{Q}.$$

The residue classes $[q_n]$ form a Cauchy sequence in $\mathbb{Q}/\mathbb{Z}$ whose image on the circle approaches $e^{2\pi i\sqrt{2}}$. That limit cannot arise from a single rational class, since $e^{2\pi i\sqrt{2}} = e^{2\pi i q}$ would imply $\sqrt{2} - q \in \mathbb{Z}$.

This construction differs from identifying the endpoints of $[0,1]$. The map $[0,1] \to S^1$, $t \mapsto e^{2\pi i t}$, realizes a *topological quotient* by gluing 0 and 1; the isometric embedding above realizes a *metric completion* by adding all the missing irrational angle classes. Both routes lead to the same familiar circle, but they answer different universal problems. That is a satisfying concrete explanation of why completion and quotient both appear in the same chapter.

§1.12 The Circle as an Endpoint Quotient

Let

$$q : [0,1] \to [0,1]/(0 \sim 1)$$

be the endpoint quotient, and define

$$F(t) = e^{2\pi i t}.$$

Since

$$F(0) = F(1),$$

the map is constant on equivalence classes. The quotient universal property produces a unique continuous map

$$\overline{F} : [0,1]/(0 \sim 1) \to S^1$$

with

$$\overline{F} \circ q = F.$$

It is bijective: every point of the circle has a parameter in $[0,1]$, and the only repeated parameter values are the identified endpoints.

The quotient of the compact interval is compact, while S^1 is Hausdorff. By the compact–Hausdorff theorem,

$$\boxed{[0,1]/(0 \sim 1) \cong S^1.}$$

A neighborhood of the glued point has preimage

$$[0, \varepsilon) \cup (1 - \varepsilon, 1],$$

so two one-sided endpoint neighborhoods become one two-sided circular neighborhood. The concrete local picture, the quotient factorization property, and the compactness theorem all describe the same gluing.